🤖 Exhaustive Algorithm Logic Manual

Complete technical documentation mapping every single Al Brooks setup modeled in the Al Engine to its exact Python formula.

Part I: The Core Engine (`algo_engine.py`)

These algorithms run natively on the central parsing engine, detecting primary structure.

1. Opening Spikes & Reversal Traps

ALGO_ENGINE.PY

BULL SPIKE | BEAR SPIKE | REVERSAL TRAP

Logic: Scans the first 12 bars (first hour). Drops strict relative size requirements on the first 4 bars (9:30 AM drop) to track absolute momentum. If a strong trend bar is immediately swallowed by a massive opposite trend bar, it's a Trap.

```
Baseline: avg_range = mean([last 20 bars range])
Threshold: 1.0x avg_range (Bars 1-4) or 1.25x avg_range (Bars 5+)
Trigger: Bar is a Strong Trend Bar AND range >= (avg_range * threshold) or (i < 5 and bar.range >= avg_range * 0.5)
```

2. Standard Breakouts

ALGO_ENGINE.PY

BULL BREAKOUT BEAR BREAKOUT

Logic: Triggers when the market forcibly breaks a localized 10-bar resistance/support wall with immense momentum, jumping the Moving Average.

```
Extreme Check: bar.close > max(past 10 highs)
Momentum Check: bar.is_strong_bull(avg_range) AND crosses EMA.
Continuation Check: Re-triggered if the previous bar was strong and the current bar is a massive anomaly: bar.range >= avg_range * 1.5
```

3. Trend Pullback Flags (H1/H2 / L1/L2)

ALGO_ENGINE.PY

H1/H2 FLAG L1/L2 FLAG

Logic: A rolling 5-bar counter that mathematically sums the number of pullback attempts. When the market breaks back in the trend direction, the counter issues the grade (H1, H2, or H3).

```
Pullback Definition: b.is_bear OR b.low < prev.low OR b.low < prev2.low
Trigger: bar.high > prev.high AND the previous sequence contained pullbacks.
```

4. Moving Average Gap Bars

ALGO_ENGINE.PY

EXHAUSTION SCALP

Logic: Fades deep exhaustion pullbacks occurring after a trend was so strong that it stayed entirely clear of the 20-EMA for at least 20 consecutive candles.

```
Constraint Check: Iterate last 20 bars: all(b.low > b.ema_20)
Trigger: The current bar finally touches: bar.low <= bar.ema_20
```

5. Wedges & Double Bottoms/Tops

ALGO_ENGINE.PY

SWING REVERSAL

Logic: Evaluates localized extremums derived from `find_swing_lows()`. A Wedge requires three monotonically ascending/descending swing points. A Double Top/Bottom requires two swing points whose absolute heights are within 0.3% mathematical variance.

```
Wedge Bottom: lo3 < lo2 < lo1 (Three consecutive lower lows in swing array)
Double Bottom: abs(lo1 - lo2) / ((lo1+lo2)/2) < 0.003
```

6. Tight Flags (ii / ioi)

ALGO_ENGINE.PY

EQUILIBRIUM BREAKOUT

Logic: Inside bars indicating extreme volatility contraction.

```
Inside Bar: b.high <= prev.high AND b.low >= prev.low
ii Trigger: Current is Inside, Previous is Inside.
ioi Trigger: Current is Inside, Previous is Outside, Prev-Prev is Inside.
```

Part II: Custom Strategy Plugins (`user_algos/`)

Advanced, specific setups parsed dynamically through the sandbox directory.

7. Major Trend Reversal (MTR)

USER_ALGOS/BEST_SETUPS.PY

TREND CHANGE

Logic: Detects the end of a dominant sequence. Requires an 80% trend adherence over 15 bars, a brutal MA violation, and a failed re-test of the absolute high/low.

```
Prior Trend: sum(1 for b in past_15 if b.close < b.ema_20) >= 12
MA Violator: any(b.close > b.ema_20 in recent_5)
Failed Test: current.low >= trend_low * 0.995 AND it forms a strong Bull signal bar.
```

8. Final Flag Reversals

USER_ALGOS/BEST_SETUPS.PY

EXHAUSTION FAILURE

Logic: A tight sideways flag in a mature trend that achieves a breakout, but instantly fails the breakout.

```
Tight Flag Check: flag_range < (bar.close * 0.015) over a 5-bar sequence.
Trigger: bar.high > flag_high (Upside breakout) AND bar.close < flag_low (Immediate catastrophic failure).
```

9. Breakout Pullbacks

USER_ALGOS/BEST_SETUPS.PY

BOPB

Logic: Validates that a massive breakout candle has successfully re-tested its origin wall and survived.

```
Breakout Candle: range > base_range * 0.5 crossing the base wall.
Trigger: Current bar pulls down exactly to the wall curr.low <= base_top AND curr.low >= (base_top * 0.998) and closes above its midpoint.
```

10. Fading Weak H1/L1 Pullbacks

USER_ALGOS/LIMIT_SETUPS.PY

LIMIT ORDER SCALP

Logic: Identifies a steep new trend and places limit orders to trap weak counter-trenders taking sub-par entries.

```
Steep Trend: ema_now > ema_10_ago * 1.002
Weak Pullback: A bear bar whose range() < ema * 0.002 (Microscopic push).
Trigger: Issues a Limit Order to BUY below the low of the weak bear bar, capturing their stop-loss failures.
```

11. Weak Breakout Tests

USER_ALGOS/LIMIT_SETUPS.PY

LIMIT ORDER TRAP

Logic: Fading breakouts that close poorly.

```
Trigger: Current bar breaks a local 5-bar extreme (e.g., low < local_low) but closes badly (close > midpoint). Issues limit order
```

```
directly on the close.
```

12. Quiet Flag MA Entries

USER_ALGOS/LIMIT_SETUPS.PY

LIMIT ORDER BASELINE

Logic: Buying the Moving Average in dead-quiet horizontal periods if the macro trend is intact.

```
Quiet Check: 4-bar average range is < ema * 0.0015
Trigger: abs(close - ema) < avg_range. Limit order issued at prior bar's extreme.
```

13. Bear Stairs Exhaustion (3rd/4th Push)

USER_ALGOS/RANGE_SETUPS.PY

REVERSAL SETUP

Logic: Counting local minima in a stair-step decline. The 3rd or 4th push usually traps late bears.

```
Counter: Increments if a bar's low is surrounded by 2 higher lows on both sides.
Trigger: minima >= 3 AND current bar is a strong bull reversal body > range * 0.5
```

14. Spike & Channel Exhaustion

USER_ALGOS/RANGE_SETUPS.PY

MOMENTUM DECAY

Logic: Compares the average range of the early phase (the spike) to the average range of the late phase (the channel) over a 20-bar lookback.

```
Trigger: If late_channel_average_range < spike_average_range * 0.5, momentum has decayed by 50%. A strong opposite close triggers the fade.
```

15. H1/L1 in Strong Fresh Spikes

USER_ALGOS/RANGE_SETUPS.PY

CONTINUATION

Logic: Entering on the very first pause of a monstrous breakout.

```
Sequence: Requires 3 consecutive strong trend bars with no overlap.
Trigger: Submits an H1 at the very first bar that prints a lower low than its predecessor but closes firmly up.
```

16. Range Boundary Fades (2nd Entries)

USER_ALGOS/RANGE_SETUPS.PY

RANGE REVERSAL

Logic: Waiting for the edge of a verified 20-bar flat box to be touched and rejected.

```
Flat Box Check: abs(ema_start - ema_end) < box_height * 0.2 (EMA is horizontal).
```

Trigger: Price pushes to `range_high * 0.998` but closes below its midpoint.